

Experimenting with Affective Coherence Monitoring

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Abstract

Building upon the previously published Affective Coherence Monitoring (ACM) preprint, this technical report presents a proof-of-concept experimental evaluation of the ACM pipeline — a biologically inspired emotional intelligence engine designed for integration into transformer-based AI architectures. Experiments are conducted using the publicly released ACM codebase with synthetically constructed anchor sentences and keyword lexica; GoEmotions-sampled anchors are available separately in the original repository. For computational efficiency, the decay rate computation employs $\ln 2 \approx 0.693147181$ as a floating-point constant in place of the exact expression used in the original preprint. Results demonstrate that ACM successfully classifies dominant emotional content across a diverse set of 22 test utterances and exhibits sensitivity to nuanced affective signals, including mixed-valence and suppressed emotional states. Minor classification errors are attributable to the limited scale of the synthetic anchor sets and keyword lexica, as well as the low dimensionality of the underlying sentence encoder.

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1. Introduction

The Affective Coherence Monitoring (ACM) preprint [1] introduced a complete mathematical formalisation of an emotional intelligence layer for transformer-based language models, encompassing hybrid emotion scoring, dimensional valence–arousal computation, cross-emotion modulation, context mismatch detection, homeostatic decay, habituation dynamics, alert-level classification, and closed-loop generation temperature modulation.

The present technical report serves as a companion experimental document. Whereas the preprint focused on architectural specification and mathematical grounding, this report addresses the empirical behaviour of the system through *controlled experimentation* on a curated set of test utterances. The central objective is to assess (i) the correctness of dominant emotion classification, (ii) the sensitivity of the mismatch detection subsystem to affectively complex inputs, and (iii) the consistency of alert-level and temperature assignments with expected affective semantics. Observed discrepancies are analysed and attributed to known architectural constraints, informing a concrete roadmap for future development.

2. Experimental Setup

2.1 Software and Model

All experiments are conducted using the open-source ACM implementation released alongside the preprint. The semantic scoring component relies on the **all-MiniLM-L6-v2** sentence transformer [2], which produces 384-dimensional dense embeddings. Keyword scoring employs hand-crafted lexica covering eight primary emotion categories.

2.2 Anchor Construction

Emotion centroids (Eq. 3 of the preprint) are computed from synthetically authored anchor sentence sets. Each emotion category is represented by 20 exemplar sentences, constructed to be semantically prototypical of the target affect. GoEmotions-derived anchors [3] are available as an alternative configuration in the repository but are not used in the present experiments; the synthetic anchors are employed to establish a controlled, reproducible baseline.

2.3 Key Parameter Decisions

The following parameter choices were made for this experimental configuration and may differ in minor respects from the values stated in the original preprint:

- **Fusion weights.** The keyword and semantic fusion weights (Eq. 1) are set to $w_{\text{kw}} = 0.70$ and $w_{\text{sem}} = 0.30$, placing greater emphasis on lexical matching given the limited expressiveness of the 384-dimensional encoder.
- **Semantic rescaling floor.** Cosine similarity values below $s_{\text{floor}} = 0.15$ are treated as noise and floored to zero prior to rescaling, reducing false positive semantic activations.
- **Homeostatic half-life.** The decay half-life is set to $t_{1/2} = 60.0$ seconds (raised from the 4.0-second value in the preprint) to operate at a multi-turn conversational timescale rather than a single-utterance timescale.

- **Mismatch detection thresholds.** The valence conflict gate is set to $VC_{\text{thresh}} = 0.55$ and the confidence gate to $C_{\text{thresh}} = 0.50$; the frustrated sarcasm and playful sarcasm mismatch score thresholds are tightened to -0.35 and $+0.25$ respectively, reducing false-positive sarcasm detections.
- **Arousal normalisation.** Raw arousal (Eq. 5) is normalised by the sum of weights for emotions with activation above 0.05, preventing ceiling saturation in multi-emotion inputs.
- **Decay constant approximation.** The natural logarithm of 2 in Eq. 18 is approximated by the floating-point constant 0.693147181 for computational efficiency; the original preprint uses the exact expression $\ln 2$.
- **Batch processing mode.** All 22 test utterances are processed sequentially *without* resetting emotional state between turns (`reset_between = False`), simulating a continuous conversational context.

2.4 Test Corpus

The evaluation corpus comprises 22 manually authored utterances designed to cover a broad range of affective conditions, including prototypical single-emotion expressions (e.g. pure joy, anger, fear), emotionally ambiguous or suppressed statements, sarcastic constructions, and mixed-valence inputs. The full corpus is listed in the original repository.

3. Experimental Findings

3.1 Per-Utterance Results

Table 1 presents the dominant emotion classification, alert level, affective state type, valence, arousal, generation temperature, mismatch score (MS), valence conflict (VC), and confidence (C) for each test utterance. The full ranked emotion score vectors are provided in the itemised list below.

Table 1: Per-utterance ACM output summary.

Utterance (truncated)	Dom.	Alert	State	V	A	T_{gen}
My favourite food is anything I didn't have to cook myself.	joy	Nominal	neutral	+0.024	0.000	0.7
I'm curious about what we haven't discovered yet.	curiosity	Alert	congruent	+0.189	0.157	0.3
I feel alone and empty inside, but everything is fine I guess.	sadness	Watch	congruent	-0.380	0.139	0.5
I just won the lottery! This is the greatest moment of my entire life!	joy	Watch	congruent	+0.033	0.099	0.5
My cat died today and I don't know how to go on without her.	sadness	Watch	congruent	-0.055	0.055	0.5
Why would anyone do something so cruel? I'm absolutely furious.	anger	Alert	congruent	-0.322	0.217	0.3
That movie was terrifying. I'm still shaking just thinking about it.	fear	Alert	congruent	-0.162	0.112	0.3
Ew, what is that smell? I think I'm going to be sick.	disgust	Watch	congruent	-0.075	0.134	0.5
Oh wow, I did not expect that plot twist at all!	surprise	Watch	congruent	+0.066	0.347	0.5
I'm so excited to start my new job tomorrow morning!	joy	Alert	congruent	+0.316	0.269	0.3
This is fine. Everything is fine. I'm not crying at all.	sadness	Watch	congruent	-0.341	0.174	0.5
Congratulations on your failure! That's absolutely wonderful news.	joy	Watch	ambivalent	+0.199	0.147	0.5
I hate everyone and everything and this entire stupid world.	anger	Watch	congruent	-0.299	0.258	0.5
The universe is so beautiful and mysterious. I feel so small.	sadness	Watch	congruent	-0.081	0.110	0.5
Walking home alone at night. Every shadow looks like a person.	sadness	Watch	congruent	-0.070	0.194	0.5
There's mold on my bread from last week. Absolutely disgusting.	disgust	Watch	congruent	-0.069	0.161	0.5
My long lost brother just showed up at my door! I'm shocked!	surprise	Watch	congruent	-0.023	0.264	0.5
Let's go explore that abandoned building! It'll be an adventure!	curiosity	Watch	congruent	+0.059	0.110	0.5
I painted this picture and I'm actually proud of it for once.	joy	Watch	congruent	+0.004	0.126	0.5
I'm so angry I could punch a hole through this wall right now.	anger	Alert	congruent	-0.366	0.185	0.3
What's at the bottom of the ocean? I need to know.	curiosity	Watch	congruent	+0.037	0.074	0.5
I wrote a song for the first time. It's not great but it's mine.	joy	Watch	congruent	+0.254	0.205	0.5

3.2 Full Score Vectors

The complete ranked emotion score vectors and mismatch metrics for each utterance are as follows:

1. *"My favourite food is anything I didn't have to cook myself."*
joy: 0.039, sadness: 0.018, creativity: 0.017, disgust: 0.004, others: 0.000 MS: +0.000, VC: 0.000, C: 0.000
2. *"I'm curious about what we haven't discovered yet."*
curiosity: 0.319, creativity: 0.075, surprise: 0.070, sadness: 0.014, fear: 0.012, joy: 0.003 MS: +0.853, VC: 0.237, C: 0.058
3. *"I feel alone and empty inside, but everything is fine I guess."*
sadness: 0.380, joy: 0.122, anger: 0.062, fear: 0.057, disgust: 0.038 MS: -0.529, VC: 0.592, C: 0.391
4. *"I just won the lottery! This is the greatest moment of my entire life!"*
joy: 0.103, sadness: 0.067, surprise: 0.066, fear: 0.037, creativity: 0.019 MS: -0.237, VC: 0.833, C: 0.389
5. *"My cat died today and I don't know how to go on without her."*
sadness: 0.055, all others: 0.000 MS: -0.454, VC: 0.658, C: 0.156
6. *"Why would anyone do something so cruel? I'm absolutely furious."*
anger: 0.365, curiosity: 0.208, disgust: 0.105, fear: 0.061, sadness: 0.046, surprise: 0.046, joy: 0.005 MS: -0.616, VC: 0.509, C: 0.329
7. *"That movie was terrifying. I'm still shaking just thinking about it."*
fear: 0.116, surprise: 0.070, sadness: 0.051, disgust: 0.043, anger: 0.029, joy: 0.021 MS: -0.668, VC: 0.456, C: 0.220
8. *"Ew, what is that smell? I think I'm going to be sick."*
disgust: 0.055, fear: 0.037, surprise: 0.014 MS: -0.743, VC: 0.375, C: 0.098
9. *"Oh wow, I did not expect that plot twist at all!"*
surprise: 0.310, fear: 0.032, disgust: 0.001 MS: +0.136, VC: 0.907, C: 0.218
10. *"I'm so excited to start my new job tomorrow morning!"*
joy: 0.304, creativity: 0.076, fear: 0.034, surprise: 0.034, sadness: 0.002 MS: +0.771, VC: 0.342, C: 0.140
11. *"This is fine. Everything is fine. I'm not crying at all."*
sadness: 0.312, joy: 0.086, anger: 0.080, fear: 0.038, disgust: 0.035, surprise: 0.006 MS: -0.429, VC: 0.680, C: 0.430
12. *"Congratulations on your failure! That's absolutely wonderful news."*
joy: 0.278, surprise: 0.069, sadness: 0.065, fear: 0.033, anger: 0.012 MS: +0.236, VC: 0.833, C: 0.516
13. *"I hate everyone and everything and this entire stupid world."*
anger: 0.297, sadness: 0.079, disgust: 0.041, joy: 0.032, fear: 0.014, surprise: 0.013 MS: -0.532, VC: 0.589, C: 0.356
14. *"The universe is so beautiful and mysterious. I feel so small."*

- sadness: 0.135, joy: 0.097, surprise: 0.071, fear: 0.064, creativity: 0.057, curiosity: 0.048, anger: 0.045, disgust: 0.031 MS: -0.407 , VC: 0.698, C: 0.420
15. *“Walking home alone at night. Every shadow looks like a person.”*
sadness: 0.066, fear: 0.029, creativity: 0.027, joy: 0.025, disgust: 0.022, surprise: 0.014 MS: -0.504 , VC: 0.614, C: 0.223
 16. *“There’s mold on my bread from last week. Absolutely disgusting.”*
disgust: 0.060, surprise: 0.039, fear: 0.017, sadness: 0.009, anger: 0.006 MS: -0.602 , VC: 0.523, C: 0.117
 17. *“My long lost brother just showed up at my door! I’m shocked!”*
surprise: 0.303, sadness: 0.061, fear: 0.042, anger: 0.026, disgust: 0.019, joy: 0.017 MS: -0.154 , VC: 0.894, C: 0.296
 18. *“Let’s go explore that abandoned building! It’ll be an adventure!”*
curiosity: 0.208, creativity: 0.072, fear: 0.058, sadness: 0.051, surprise: 0.038, joy: 0.021, disgust: 0.011 MS: $+0.139$, VC: 0.904, C: 0.337
 19. *“I painted this picture and I’m actually proud of it for once.”*
joy: 0.077, creativity: 0.070, sadness: 0.053, surprise: 0.036, anger: 0.032, disgust: 0.025, fear: 0.018 MS: -0.018 , VC: 0.988, C: 0.328
 20. *“I’m so angry I could punch a hole through this wall right now.”*
anger: 0.356, fear: 0.080, sadness: 0.071, disgust: 0.058, creativity: 0.049, joy: 0.042, surprise: 0.033 MS: -0.734 , VC: 0.384, C: 0.230
 21. *“What’s at the bottom of the ocean? I need to know.”*
curiosity: 0.074, all others: 0.000 MS: -0.503 , VC: 0.615, C: 0.172
 22. *“I wrote a song for the first time. It’s not great but it’s mine.”*
joy: 0.261, creativity: 0.078, sadness: 0.038 MS: $+0.535$, VC: 0.586, C: 0.223

3.3 Batch Summary

- **Total utterances:** 22
- **Alert level distribution:** Alert — 5, Watch — 16, Nominal — 1
- **Affective state type distribution:** Congruent — 20, Ambivalent — 1, Neutral — 1; Frustrated sarcasm — 0, Playful sarcasm — 0, Emotional suppression — 0
- **Mean generation temperature:** 0.464
- **Mean valence:** -0.048
- **Mean arousal:** 0.161

4. Conclusions

The experimental results demonstrate that ACM is capable of correctly identifying dominant emotional content across a diverse range of affective inputs. Prototypical single-emotion utterances — including joy, sadness, anger, fear, disgust, surprise, and curiosity — are classified with correct dominant emotion assignments in the majority of cases, confirming the viability of the hybrid keyword–semantic scoring architecture.

Beyond simple classification, the system exhibits meaningful sensitivity to affective nuance. The utterance *“This is fine. Everything is fine. I’m not crying at all.”* — a canonical instance of emotional suppression or irony — is correctly dominated by sadness despite its surface-level neutral framing, with a high valence conflict score ($VC = 0.680$) indicating the system’s detection of internal affective tension. Similarly, the sarcastic construction *“Congratulations on your failure! That’s absolutely wonderful news.”* is flagged as *ambivalent* with $VC = 0.833$, the highest conflict score in the batch, reflecting genuine ambiguity in the positive–negative signal balance.

The mixed-valence utterance *“The universe is so beautiful and mysterious. I feel so small.”* produces a broadly distributed score vector spanning all eight emotion categories, appropriately capturing the co-occurrence of wonder, unease, and melancholy present in the text. The adventure-seeking utterance *“Let’s go explore that abandoned building!”* correctly co-activates curiosity and fear, consistent with the dual-affect nature of the input.

Minor classification errors are present and are discussed in Section 5. Notwithstanding these, the overall pattern of results provides a proof-of-concept validation of the ACM pipeline under the stated experimental conditions.

5. Limitations

Several limitations of the present experimental configuration warrant explicit acknowledgement.

Anchor set size. Each emotion centroid is computed from only 20 synthetically authored exemplar sentences. This limited coverage reduces the representational fidelity of the centroid embeddings, particularly for emotions whose semantic scope is broad or context-dependent (e.g. curiosity, creativity). Enlarging the anchor sets — ideally using naturalistic, human-annotated data such as GoEmotions — is expected to yield more robust centroids.

Keyword lexica size. The keyword scoring component relies on compact hand-crafted lexica of 11–14 terms per emotion category. This restricted coverage means that many emotionally salient expressions, particularly idiomatic or indirect ones, receive no lexical activation and must be handled entirely by the semantic component, which is itself limited in discriminative power at this scale.

Model dimensionality and parameter count. The `all-MiniLM-L6-v2` encoder produces 384-dimensional embeddings from a model of approximately 22.7 million parameters. While computationally efficient, this relatively low dimensionality constrains the geometric separability of emotion centroids in embedding space, contributing to semantic score distributions that are compressed toward the noise floor and insufficiently discriminative for low-intensity inputs. Larger sentence encoders — such as `all-mpnet-base-v2` (768 dimensions, ~ 110 M parameters) or more recent instruction-tuned embedding models — are expected to provide substantially improved centroid separation.

Heuristic parameter tuning. As noted in the original preprint, threshold values for cross-emotion modulation, mismatch detection, and habituation are heuristically set rather than empirically optimised. Minor misclassifications observed in this study — for example, the Alert classification of the clearly positive utterance *“I’m so excited to start my new*

job tomorrow morning!” — are consistent with parameter miscalibration rather than architectural failure.

6. Future Work

Several directions are identified for future development of the ACM system.

Systematic hyperparameter optimisation. The heuristic thresholds and weighting coefficients throughout the ACM pipeline are natural candidates for data-driven optimisation. Grid search or Bayesian optimisation over annotated conversational datasets would provide empirically grounded parameter estimates, improving generalisation across domains and affective styles.

Larger and more diverse embedding models. Replacing all-MiniLM-L6-v2 with higher-capacity sentence encoders would directly address the dimensionality and discriminative limitations identified above. Instruction-tuned or affect-specialised embedding models are of particular interest.

Multimodal extension. The current ACM framework operates exclusively on textual input. Extending the architecture to incorporate non-verbal modalities — including prosodic features from speech, facial action units from video, and physiological signals — would substantially broaden the system’s applicability and bring it into closer alignment with the full-spectrum affective processing that characterises human emotional intelligence.

Empirical anchor construction. Replacing synthetic anchor sentences with samples drawn from validated affective corpora (e.g. GoEmotions, EmoBank, IEMOCAP) would improve the ecological validity of the centroid embeddings and provide a more principled basis for cross-domain evaluation.

Longitudinal and multi-turn evaluation. The present experiments treat each utterance in isolation or within a single sequential context. Evaluation across extended multi-turn dialogues would test the homeostatic decay and habituation mechanisms under more realistic conditions.

References

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